



Characterization of polyoxyalkylene block copolymers by combination of different chromatographic techniques and MALDI-TOF-MS

Muhammad Imran Malik^{a,b}, Bernd Trathnigg^{a,b,*}, Karin Bartl^{b,c}, Robert Saf^{b,c}

^a Institute of Chemistry, Karl-Franzens-University, Heinrichstrasse 28, A-8010 Graz, Austria

^b NAWI Graz, Central Polymer Laboratories (CePoL), Heinrichstrasse 28 and Stremayrgasse 16, A-8010 Graz, Austria

^c Institute for Chemistry and Technology of Materials (ICTM), Graz University of Technology, Stremayrgasse 16, A-8010 Graz, Austria

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ABSTRACT

Polyoxyalkylene diblock copolymers (consisting of PEO as hydrophilic block and PBO or PHO as hydrophobic block) are characterized by combination of two dimensional liquid chromatography and MALDI-TOF-MS. Liquid chromatography under critical conditions (LCCC) is used as first dimension and fractions are collected, mobile phase evaporated and diluted in the mobile phase used in the second dimension (SEC, LCCC or LAC). This two-dimensional chromatography in combination of MALDI-TOF-MS gives information about purity of reaction products, presence of the byproducts, chemical composition and molar mass distribution of all the products.

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1. Introduction

Polyoxyalkylene block copolymers are a versatile class of polymers. Depending on the substituent on the oxirane ring, these polymers show very different solution behaviour. Polyoxyethylenes (without any substituent) are very hydrophilic, while polymers with methyl, ethyl and butyl substituent are more lipophilic in the same order. Hence polyoxyalkylene block copolymers of ethylene oxide (EO) and higher alkylene oxides (e.g. propylene oxide (PO), butene oxide (BO) and hexene oxide (HO)) are amphiphilic in nature. These type of copolymers were developed and commercialized some 50 years ago and are still being used in many different fields like cosmetics, medicines, cleaners, lubricants, and many others [1]. Poly(ethylene oxide) (PEO) is used as hydrophilic block while polymers of higher alkylene oxides like poly(propylene oxide) (PPO), poly(butene oxide) (PBO) and poly(hexene oxide) (PHO) are used as hydrophobic moiety. Most commercialized products of this class are poloxamers (PEO–PPO–PEO) [2,3–16]. There are also many reports on EO–PO diblock copolymers [1,5,17–20], but literature is very limited on

higher polyoxyalkylene block copolymers [7,21–25]. However, they may have very special physical properties due to more hydrophobicity of BO or HO block [26].

As polyoxyalkylene block copolymers are often synthesized by anionic ring opening polymerization of alkylene oxides, one has to take into account possible side reactions leading to the formation of homopolymers. In the presence of water polyalkylene glycols may be formed [21,23], which can be avoided by drying the starting materials carefully. Another side reaction can, however, not completely be avoided: chain transfer to the monomer leads to homopolymers with unsaturated end groups. This reaction has mainly been studied for PO [20,27–29], but it may also happen in the polymerization of higher alkylene oxides. One may speculate that it may be less important in this case because of steric hindrance on the methylene group next to the oxirane ring.

The lengths of homopolymer diol, homopolymer with unsaturated end group and higher alkylene oxide block in block copolymer may be very different. One may expect that the diol has the highest molar mass, as it may grow in both directions, and starts to grow at the beginning of the polymerization. The unsaturated homopolymer, however, must have a much lower molar mass, as the chain transfer happens all the time, consequently some new chains will still be started at rather high conversion of the monomer.

As the amount of these homopolymers, their MMD, and functionality can strongly affect the properties of the product; reliable analysis methods are required, which yield information on molar

* Corresponding author at: Institute of Chemistry, Karl-Franzens-University, Heinrichstrasse 28, A-8010 Graz, Austria. Tel.: +43 316 3805328; fax: +43 3163809840.

E-mail address: bernd.trathnigg@uni-graz.at (B. Trathnigg).

mass distributions, content and molar mass of homopolymer and the length of individual blocks in the copolymer. Such a full characterization should reveal the source of side reactions and indicate ways to avoid or at least minimize the formation of side products.

Obviously, simple methods cannot provide all these information. This can only be achieved by an appropriate combination of different techniques, such as two-dimensional liquid chromatography (2D-LC). The hyphenation of HPLC with MALDI-TOF-MS, FTIR-spectroscopy and NMR are also possible [30–45].

In our recent papers about polyoxyalkylene block copolymers [24,25], it was illustrated how SEC (with dual detection) and LCCC could be used to analyze polyoxyalkylene block copolymers. In this study, we have now hyphenated LCCC with SEC, LAC, LCCC and MALDI-TOF-MS, and compare the results obtained by these approaches.

2. Experimental

Higher alkylene oxides (butene oxide and hexene oxide) and polyethylene glycol monomethyl ethers (MeO-PEO) with different molecular weights (550, 1100, 2000) were purchased from Fluka (Buchs, Switzerland) and sodium hydride (60% dispersion in mineral oil) from Aldrich. MeO-PEO was dried by azeotropic distillation with toluene. Alkylene oxides were refluxed over Calcium hydride (Fluka).

Block copolymers were synthesized by microwave-assisted anionic ring opening polymerizations of higher alkylene oxides (butene oxide, hexene oxide) using MeO-PEO as initiator and sodium hydride as co-initiator according to the literature [24,25]. Briefly all reagents (MeO-PEO, alkylene oxide monomer and sodium hydride) were taken in 2–5 mL (filling volume) microwave process vials (Biotage AB) and sealed microwave vials were introduced into the cavity of single-mode microwave reactor (Biotage Initiator Eight EXP 2.0, absorbance level: high). Polymerizations were carried out at 160 °C till complete conversion. After polymerization, equimolar quantity of acetic acid was added to neutralize the active chain end.

All separations were performed using the density detection system DDS 70 (CHROMTECH, Graz, Austria) which has been developed by our group. Data acquisition and processing was carried out by using software package CHROMA.

In the semipreparative LCCC system, which was used in the first dimension, the mobile phase was delivered by JASCO 880 PU pump (Japan Spectroscopic Company, Tokyo, Japan) at a flow rate of 2.0 mL min^{-1} . Samples were injected manually using Rheodyne 7125 injection valve (Rheodyne, Cotati, CA, USA) equipped with a $500 \mu\text{L}$ loop. A semi-preparative column was used for all measurements: Spherisorb ODS2 ($250 \text{ mm} \times 10 \text{ mm}$, $5 \mu\text{m}$, $80 \text{ \AA}$) from PhaseSep (Deeside, UK). A SICON LCD 201 RI detector was combined with the DDS70. Fractions were collected using an Advantec SF 2120 fraction collector (Advantec, Dublin, CA, USA). Critical conditions for PBO were realized in 88 wt% THF–water and for PHO in 92 wt% THF–water.

The solvent was removed from the fractions by freeze drying using a Benchtop 2K ES (Virits, USA). The limits of the instrument are -60°C and 2.6 Pa. Samples were frozen to about -40°C and were subjected for drying to benchtop 2K ES which was pre-maintained at a temperature below -50°C and a pressure below 3 Pa. Typically it took 2–3 h for complete drying of the samples.

In the SEC system (second dimension), we used a modular system consisting of a Gynkotek 300C pump, a VICI injector equipped with a $100 \mu\text{L}$ sample loop, two column selection valves Rheodyne 7060 (Rheodyne, Cotati, CA, USA), a density detec-

tion system DDS 70 (Chromtech, Graz, Austria) coupled with a Bischoff 8110 refractive index detector (Bischoff, Leonberg, Germany). SEC measurements were performed on a $7.8 \text{ mm} \times 600 \text{ mm}$ Phenogel 100 Å column (Phenomenex, Torrance, CA, USA) or a $7.8 \text{ mm} \times 300 \text{ mm}$ PLgel MIXED E column (Polymer Laboratories, Church Stretton, Shropshire, UK) in chloroform at a flow-rate of 1.00 mL min^{-1} . Sample concentrations were $3.0\text{--}10.0 \text{ g L}^{-1}$. The columns were calibrated with PPGs from Sigma–Aldrich and FLUKA (Buchs, Switzerland). Molar mass distributions were calculated using CHROMA.

In the analytical LCCC system (second dimension), the mobile phase was delivered by an ISCO 2350 HPLC pump (ISCO, Lincoln, NE, USA) at a flow rate of 0.5 mL min^{-1} . Samples were injected manually using a Rheodyne 7125 injection valve (Rheodyne, Cotati, CA, USA) equipped with a $50 \mu\text{L}$ loop. A refractive index detector RI 2414 (Waters) or a SEDEX 45 ELSD (Sedere, France) was connected to the DDS 70. Nitrogen was used as carrier gas, and the pressure at the nebulizer was set to 1.0 bar (evaporator temperature: 30°C). Normal phase column used for LCCC in second dimension was Nucleosil 100-5 OH $5 \mu\text{m}$, silica-based diol phase; $4.6 \text{ mm} \times 250 \text{ mm}$; particle diameter: $5 \mu\text{m}$; pore diameter: $100 \text{ \AA}$ (Macherey-Nagel, Dueren, Germany). Critical conditions for PEO exist at 75.7 wt% acetone–water.

In LAC, the separations in the second dimension were performed on a gradient system Ultimate 3000, consisting of a pump DGP-3600A, solvent degasser SRD-3600, column thermostat TCC-3000, autosampler WPS-3000SL, all from Dionex (Germerink, Germany), an evaporative light scattering detector PL-ELS 2100 (Polymer Laboratories, Church Stretton, Shropshire, UK). Data acquisition and processing was performed using the software Chromeleon (Dionex, Germerink, Germany). Gradient separations were performed on a Nucleosil 4000-7 C18 (Macherey-Nagel, Germany); silica-based octadecyl phase; $4 \text{ mm} \times 250 \text{ mm}$; particle diameter = $7 \mu\text{m}$; pore size = $4000 \text{ \AA}$ at a flow rate of 0.5 mL min^{-1} . The injected sample volumes depended on the total volume of the fraction from the first dimension in order to inject comparable sample sizes.

The following gradient was used: start 80 vol% methanol–water, in 10 min to 90 vol% methanol, in 30 min 95 vol% methanol, in 40 min 98 vol% methanol, 15 min constant at 98 vol%, then back to 80 vol% methanol in 1 min and 4 min at 80 vol% methanol.

The solvents for HPLC (THF, chloroform, acetone and water, all HPLC grade) were purchased from Roth (Karlsruhe, Germany).

Mobile phases for LCCC (both for analytical and preparative scale) were mixed by mass and vacuum degassed, their composition was controlled by density measurement using a DMA 60 density meter equipped with a measuring cell DMA 602M (A. Paar, Graz, Austria).

The columns and density cells in all systems were placed in thermostatted boxes, in which a temperature (25.0°C for LCCC and at 30.0°C for SEC) was maintained for all measurements using a thermostat Lauda RM6, Lauda-Königshofen, Germany).

MALDI-TOF Mass Spectrometry was performed on a Micromass TofSpec 2E Time-of-Flight Mass Spectrometer. The instrument is equipped with a nitrogen laser (337 nm wavelength, operated at a frequency of 5 Hz), and a time lag focusing unit. Ions were generated by irradiation slightly above the threshold laser power. Positive ion spectra were recorded in reflectron mode applying an accelerating voltage of 20 kV and externally calibrated with a suitable mixture of poly(ethyleneglycol)s (PEO). The spectra of 100–150 shots were averaged to improve the signal-to-noise ratio. Analysis of data was done with MassLynx-Software V3.5 (Micromass/Waters, Manchester, UK).

Sample were prepared by mixing a solution of dithranol ($c = 10 \text{ mg mL}^{-1}$ in THF), a solution of the analyte ($c = 2 \text{ mg mL}^{-1}$ in THF), and a solution of CF_3COONa (NaTFA; $c = 0.1 \text{ mg mL}^{-1}$ in THF), in a ratio of 7:1:1 (v/v/v). $0.5 \mu\text{L}$ of the resulting mixture were

deposited on the sample plate (stainless steel) and allowed to dry under air.

3. Results and discussion

3.1. HPLC of polymers

Functional polymers and oligomers can be characterized by using different chromatographic techniques, which separate according to different criteria. The type of chromatographic separation is determined by the so-called interaction parameter c , which describes the interaction of a structural unit with the stationary phase [46].

This parameter is negative in size exclusion chromatography (SEC) and positive in liquid adsorption chromatography (LAC). At the critical adsorption point (CAP) the interaction parameter equals zero, and the corresponding block becomes “chromatographically invisible” [47–50]. This effect is utilized in liquid chromatography under critical conditions (LCCC), which allows the separation of a polymer homologous series according to their end groups [48,51–53].

In the case of amphiphilic polymers the interaction parameters c_A and c_B of the individual structural units (A and B) may have different values. If both c_A and c_B are negative, the molar mass distribution (MMD) can be determined by size exclusion chromatography (SEC), with retention decreasing with increasing molar mass, and only a minor influence of chemical composition. SEC yields the overall MMD, with dual detection [54–56] one may obtain information on the chemical composition along the MMD [56–58]. SEC is, however, not capable of discriminating a mixture of block copolymers and homopolymers. In the case of higher polymers, gradient LAC [59–66] or LC under limiting conditions [67–69] can be applied. In the range of lower molecular mass, however, LCCC is the best approach to separate homopolymers from the block copolymer. In AB diblock copolymers (e.g. A = EO and B = PO or another alkylene oxide) one may use the following strategies: At the CAP for the unit A, a separation according to the block B may be achieved by a SEC or a LAC mechanism, while at the CAP for the block B, the length of the block A may be obtained in the same way. The application of these methods to EO–PO block copolymers has been described recently [20,45,70–72].

3.2. Analysis of the raw samples by SEC with dual detection

As has been shown previously, SEC with coupled density and RI detection allows the determination of chemical composition

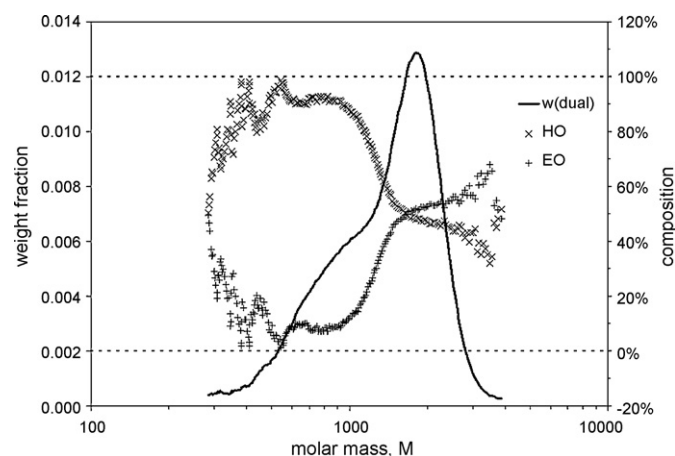


Fig. 1. Chemical composition across MMD of the MeO-PEO1100-b-PHO raw sample, as obtained from SEC with dual detection.

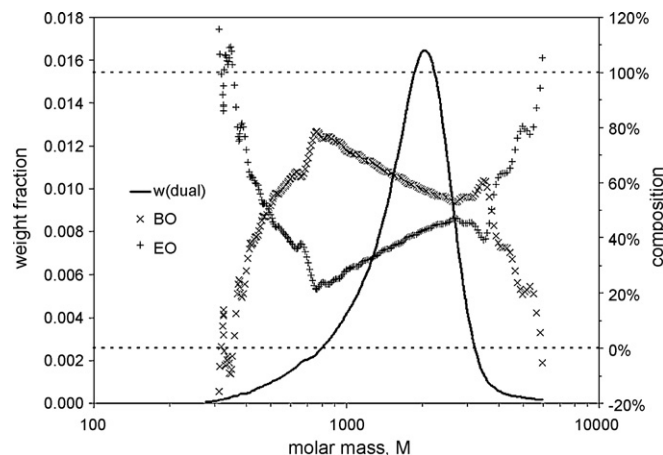


Fig. 2. Chemical composition across MMD of the MeO-PEO1100-b-PBO raw sample, as obtained from SEC with dual detection.

along the MMD. Fig. 1 depicts that MeO-PEO1100-b-PHO sample contains a low molecular shoulder, which evidently contains the homopolymer of HO. In the MMD of MeO-PEO1100-b-PBO there is no pronounced shoulder, but the BO content decreases with increasing molar mass, which indicates, that this sample also contains some homopolymer of BO (see Fig. 2).

The main part of the MMD appears to be the copolymer, but it still could contain some amount of homopolymers with comparable molar mass. Therefore it was necessary to separate these samples by LCCC on a reversed phase column at the CAP for PBO or PHO, respectively. Under such conditions, the block copolymer (and the initiator) would elute in SEC mode and could thus be separated from PBO and PHO.

3.3. Semipreparative LCCC on a reversed phase column in the first dimension

Recently it was reported that there exists a CAP for BO and HO on reverse phase column (Jordi Gel DVB 500 RP) packed with a stationary phase consisting of 100% poly(divinyl benzene) in THF: water mobile phases containing 85% and 87% THF, respectively [24]. On most common octadecyl reverse phase columns, it could also be realized with a slightly higher content of THF (87–92% for butene oxide, 90–95% for hexene oxide) [73]. Under the conditions mentioned above, the hydrophilic PEO as well as the block copolymer elute in SEC order. If there is any hydrophobic homopolymer formed during synthesis, it should elute at the

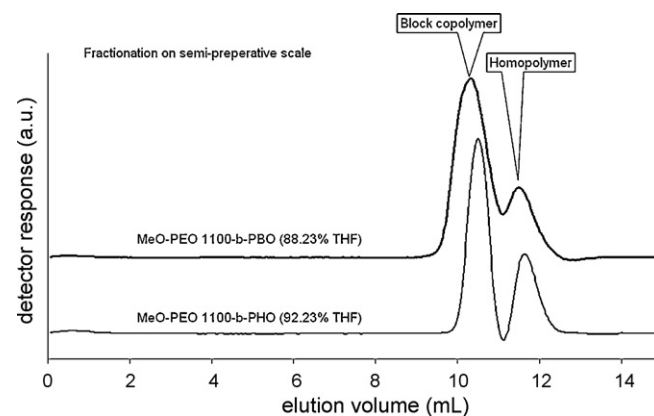


Fig. 3. LCCC of MeO-PEO1100-b-PBO in 88.23 wt% THF (CAP of PBO) and MeO-PEO1100-b-PHO in 92.23 wt% THF (CAP of PHO) (Spherisorb ODS2 prep., detection: density).

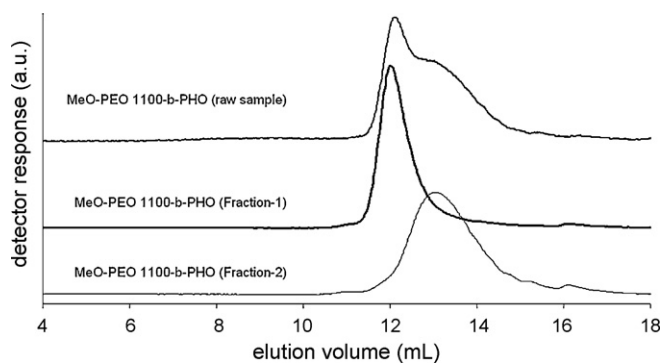


Fig. 4. Overlay of SEC chromatograms of the MeO-PEO1100-b-PHO raw sample and the fractions 1 and 2 from LCCC (from Fig. 3). Detection: density.

void volume of column. In this study we used an octadecyl column with 10 mm inner diameter for the semipreparative separation of block copolymer from homopolymer. As shown in Fig. 3, the block copolymer (fraction 1) can be reasonably separated from the homopolymer (fraction 2). The fractions from LCCC were collected and the mobile phase was evaporated by high vacuum freeze drying. The dried fractions were subsequently analyzed by SEC, LCCC, LAC and MALDI-TOF-MS.

3.4. Size exclusion chromatography in second dimension

After evaporating the mobile phase from the LCCC separation, the fractions were dissolved in chloroform (used as mobile phase for SEC) and analyzed by SEC. Figs. 4 and 5 show the density trace of the SEC chromatograms obtained with the raw samples and the fractions from LCCC, respectively (from Fig. 3). Fraction 1 should be the block copolymer while fraction 2 should be the homopolymer (although there can be different functionalities of homopolymer).

Obviously, the homopolymer fraction represents the low molecular weight shoulder in the chromatogram of the raw samples, while the copolymer fraction contains the higher molecular weight end of the original MMD.

While the homopolymer fraction could be well separated in the block copolymer with HO, the MMDs of both fraction overlap in the sample with BO. Anyway, dual detector SEC shows clearly, that the composition of the copolymer fraction is constant (with only a minor low molecular part of a different composition) and fraction 2 really contains the homopolymer.

As can be seen in Fig. 6, fraction 1 (from Fig. 3) consisted of the block copolymer, which exhibited a maximum of the MMD slightly above 2000 with comparatively narrow distribution and contained about 55% BO (see Fig. 3). This is reasonable keeping in mind that

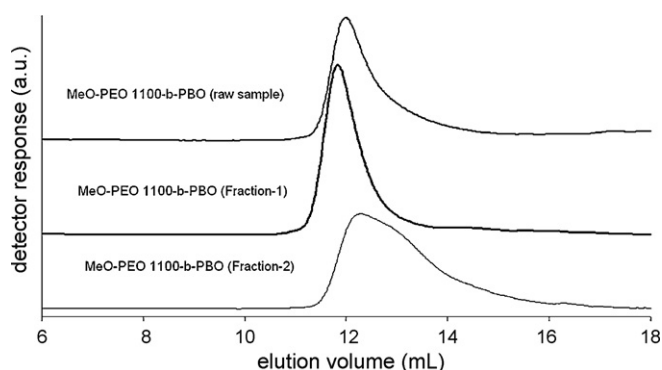


Fig. 5. Overlay of SEC chromatograms of the MeO-PEO1100-b-PBO raw sample and the fractions 1 and 2 from LCCC (from Fig. 3). Detection: density.

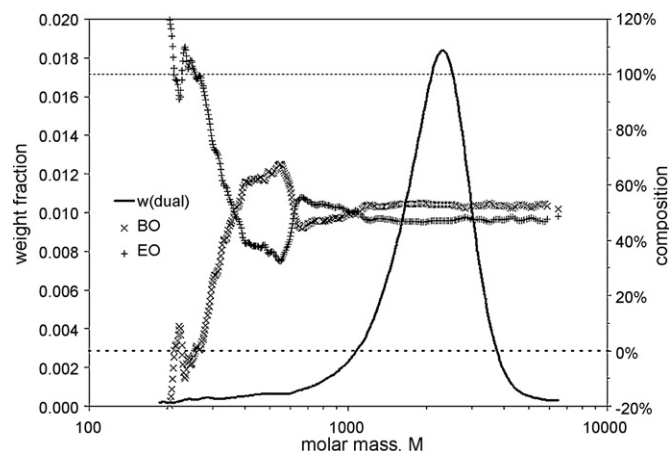


Fig. 6. Chemical composition across MMD of fraction 1 (MeO-PEO1100-b-PBO, from Fig. 3), as obtained from SEC with dual detection.

some of the monomer (from the initial ratio of 1:2) went to the PBO homopolymer. If both blocks have approximately the same length and a comparable polydispersity, one may expect a constant composition over the entire MMD, which is indeed the case (Fig. 6). Fraction 2 (from Fig. 3) had a very broad MMD with a maximum below 2000 and a BO content of almost 100% (see Fig. 7). Note that the low molecular end of the MMD has a lower molar mass than the starting material (MeO-PEO1100).

3.5. LCCC on a normal phase column in the second dimension

From the results shown above it is not clear, whether the copolymer fractions (fraction 1) (from Fig. 3) contained some residual MeO-PEO homopolymer (which is used as initiator), which should coelute with the block copolymer in LCCC on a reversed phase column. Thus, the fractions 1 were analyzed in the second dimension at the CAP for EO on a normal phase column.

Under these conditions, the products containing hydrophobic moieties (PBO and PHO homopolymers and copolymer containing BO and HO units, respectively) elutes in SEC mode [73]. Hence the block copolymer should be separated from PEO or MeO-PEO, which would elute at the void volume, and could thus be detected. The results of these investigations (Fig. 8) indicate, that there were no significant amounts of unreacted MeO-PEO in the copolymer fraction from the first dimension.

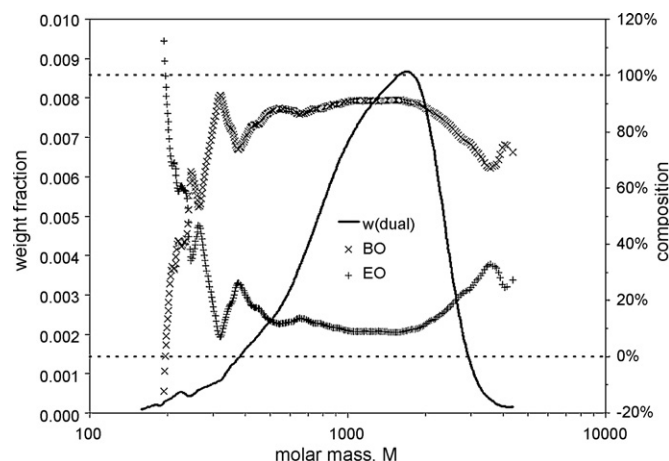


Fig. 7. Chemical composition across MMD of fraction 2 (PBO homopolymer, from Fig. 3), as obtained from SEC with dual detection.

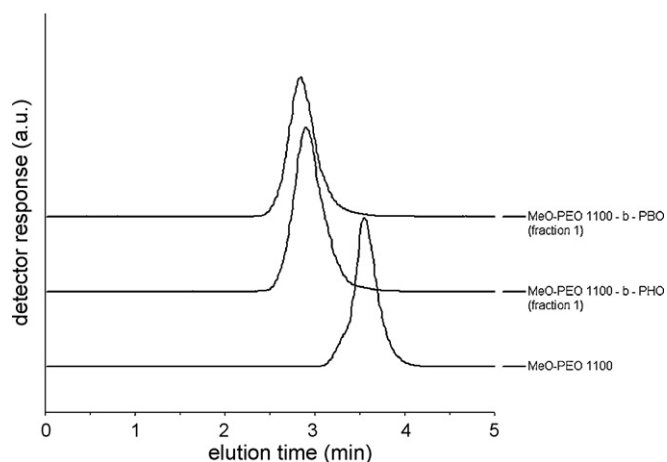


Fig. 8. Separation of block copolymers (fraction 1, from Fig. 3); from PEO at CAP of PEO and exclusion for hydrophobic moiety.

3.6. LAC in second dimension

Liquid adsorption chromatography is powerful technique for the separation of oligomers of low molecular weight samples. Using methanol–water as mobile phase on a octadecyl RP columns, there exists a CAP for EO for a wide range of mobile phase composition (75–95% methanol). In the range mentioned above EO is fairly critical. Consequently, a gradient (see Section 2) was run in this range to achieve the separation of according oligomers of BO. It is clear from Fig. 9 that the raw sample contained two series of oligomers, one with quite low molecular weight while second had higher molecular weight. From the LAC-trace of fraction 1 (from Fig. 3) it can be deduced that this is indeed a block copolymer with comparatively higher molecular weight. Fraction 2 (from Fig. 3) exhibited a very broad distribution having a low molecular maximum and a second maximum at higher molecular weight. The lower molecular weight part of fraction 2 has two series. One is assigned to diols while the other is interpreted as a series of oligomers with unsaturated end groups. The higher molecular weight part of fraction 2 contained only one series (results are similar to that of spectroscopy, see below). This may be due to different sources of side reactions

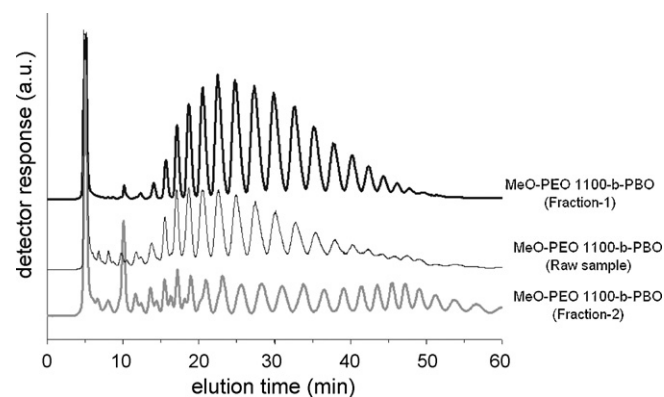


Fig. 9. LAC separation of MeO-PEO1100-b-PBO and its fractions (from Fig. 3); according to BO block on CAP of PEO (gradient: see text).

as explained in our previous paper [45]. The evaporative light scattering detector (ELSD) was used for LAC in second dimension. This detector can only detect non-volatile component of analyte. As discussed above the conditions used for LAC separation are critical for PEO. Therefore if there is some unreacted initiator is present in the sample, it would elute at the void volume of column (which should not give any signal by ELSD otherwise). It can be noted that there are some traces of unreacted MeO-PEO in the sample that were not detected by LCCC on normal phase column in second dimension. This agrees quite well with the results from spectroscopy.

For the block copolymers containing polyhexene oxide as hydrophobic moiety, such a separation was not possible since the interaction parameter of hexene oxide in such a mobile phase composition is too high. Experiments to find a system with CAP for EO and reasonable interaction parameter for hexene oxide (2–3) with good solubility were not successful until yet.

3.7. MALDI-TOF-MS of the raw samples and in the second dimension

To verify the results from the chromatographic experiments MALDI-TOF-MS investigations were performed. MALDI-MS is well known as very powerful technique for samples with low polydispersity. If, however, the polydispersity and/or heterogeneity

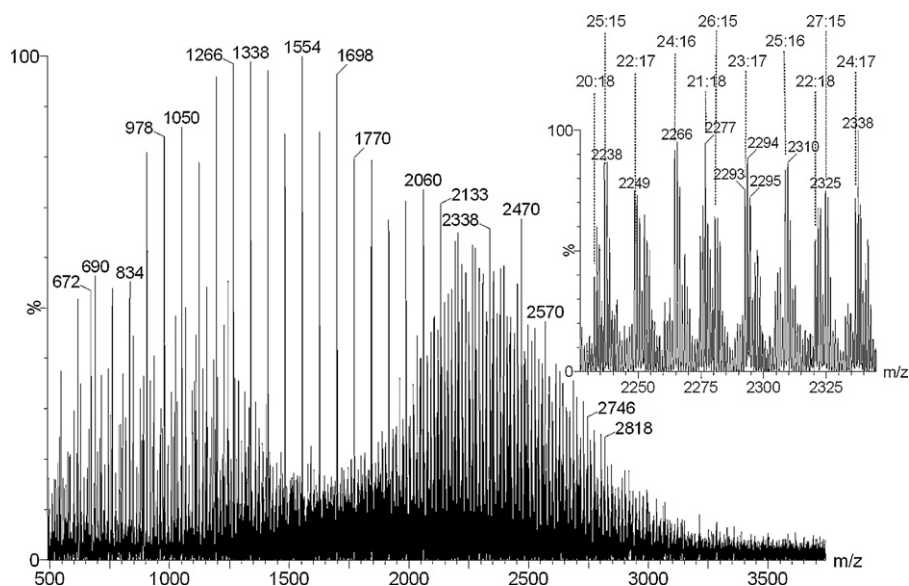


Fig. 10. MALDI-TOF mass spectrum of the MeO-PEO1100-b-PBO raw sample showing the block copolymer (1500–3300 Da), and the PBO homopolymer (intense series between 750 and 2500 Da). Inset: Expanded view of the mass range 2225–2350 Da. The labels n:m indicate the monoisotopic mass of copolymer ions $[\text{MeO}(\text{EO})_n-(\text{BO})_m-\text{H}+\text{Na}]^+$.

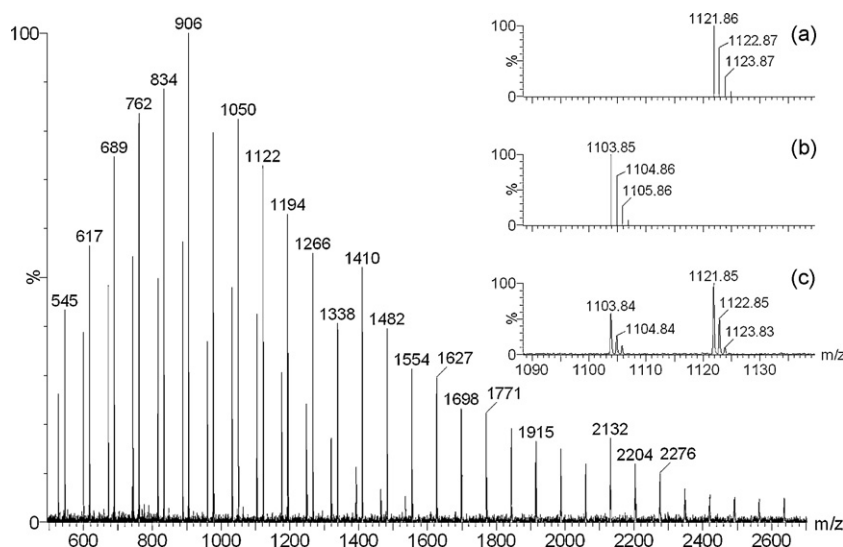


Fig. 11. MALDI-TOF mass spectrum of fraction 2 (from Fig. 3); showing the PBO-diol homopolymer (major series) as well as PBO with one unsaturated endgroup (PBO-H₂O, minor series). Inset: Comparison of the theoretical isotope pattern of [PBO_{n=15}-diol + Na]⁺ (C₆₀H₁₂₂O₁₆Na) (a), [PBO_{n=15}-H₂O + Na]⁺ (C₆₀H₁₂₀O₁₅Na) (b), and the experimental result (c).

increase, data should be handled with great care since phenomena like mass discrimination can have a significant impact on the observable results. In this study, the raw samples as well as the fractions from LCCC (from Fig. 3) were investigated.

The MALDI spectrum of the MeO-PEO1100-b-PBO raw sample is reproduced in Fig. 10. In principle, the spectrum shows all important components detected with the chromatographic techniques.

The copolymer MeO-PEO1100-b-PBO was observed as a series of ions [MNa]⁺ between 1500 and 3300 Da, with the maximum of the MMD at about 2300 Da, which fits well to the data obtained by chromatography (Figs. 5 and 6). Due to the polydispersity of both blocks this part of the spectrum appears complex, but all important signals can be assigned to [MNa]⁺ ions of corresponding copolymer macromolecules as shown for selected examples in the inset of Fig. 10. The relative intensity of the assigned signals can be used for a simple estimation of the average EO and BO content of the copolymer. For example, a relative intensity of 75% was observed for [MeO(EO)_n-(BO)_m-H+Na]⁺ with $n=22/m=17$, 92%

for $n=24/m=16$, 94% for 21/18, 62% for 26/15, and 74% for 23/17, respectively (compare inset of Fig. 10). The average ratio of EO/BO repeating units calculated from these values is 1.00/0.72. Multiplication with the mass of the corresponding repeating unit and normalization to 100% gives an EO/BO mass ratio of 45.9/54.1. Note that this value fits excellent to the BO content of 55% determined via chromatography.

It has to be mentioned, that the spectrum in Fig. 10 was recorded with instrumental settings (e.g. laser power) optimised for the investigation of the copolymer. Consequently, the data obtained in the lower mass range have to be treated with care, especially with respect to MMDs observed. Nevertheless, the MALDI spectrum of the raw sample clearly show the presence of the PBO diol homopolymer, which was observed as a series of ions [MNa]⁺ too (e.g. at $m/z=1266$, 1338, 1410, 1482, 1554 Da). Below 1500 Da the spectrum appears again complex due to several interfering series of ions observed with reduced intensity. In this mass range the unsaturated PBO homopolymer (compare the more detailed discussion

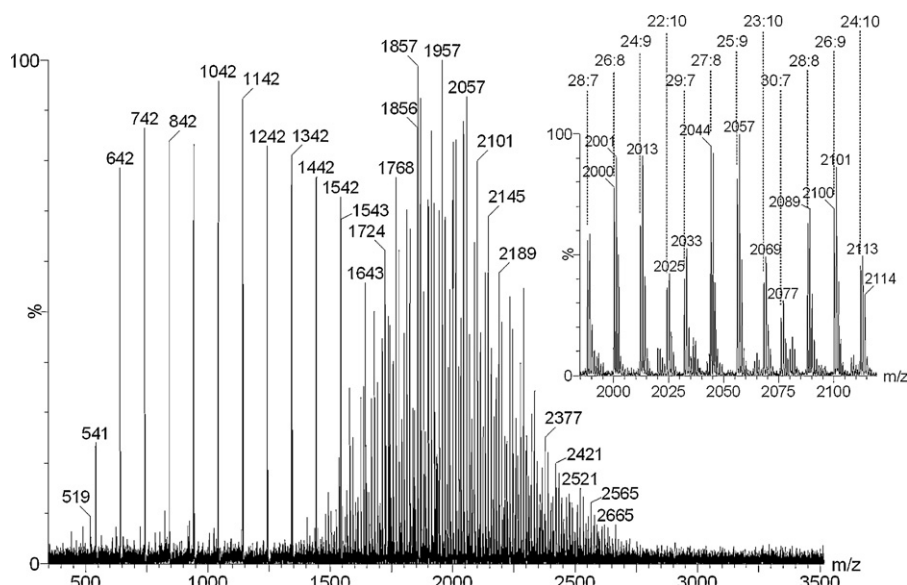


Fig. 12. MALDI-TOF mass spectrum of the MeO-PEO1100-b-PHO raw sample showing the block copolymer (1300–2800 Da) as well as the PHO homopolymer (500–2000 Da). Inset: Expanded view of the mass range 1985–2120 Da. The labels $n:m$ indicate the monoisotopic mass of copolymer ions [MeO(EO)_n-(HO)_m-H+Na]⁺.

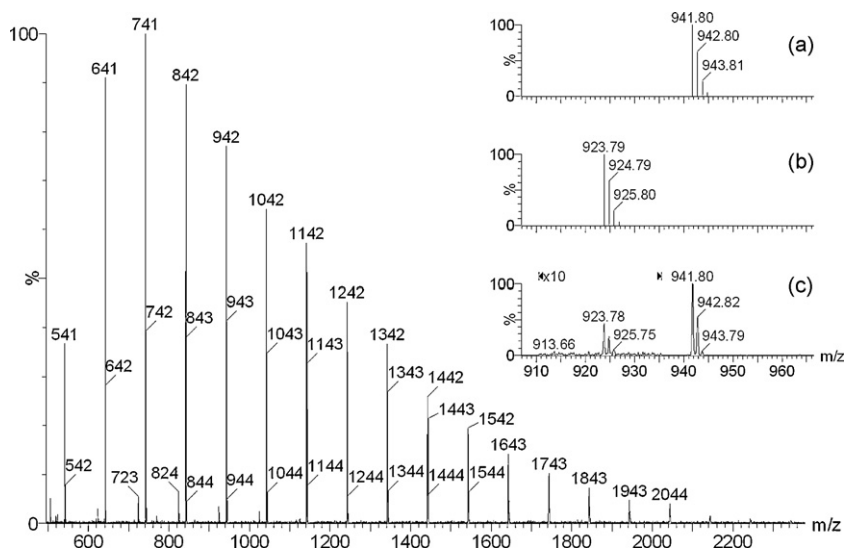


Fig. 13. MALDI-TOF mass spectrum of fraction 2 (from Fig. 3); showing the PHO-diol homopolymer and with low intensity PHO with one unsaturated endgroup (PHO-H₂O). Inset: Comparison of the theoretical isotope pattern of [PHO_{n-9}-diol+Na]⁺ (C₅₄H₁₁₀O₁₀Na) (a), [PHO_{n-9}-H₂O+Na]⁺ (C₅₄H₁₀₈O₉Na) (b), and the experimental result (c).

below) and a small residual amount of the MeO-PEO1100 initiator were detected as [MNa]⁺ ions, as well as some matrix clusters. The agreement between results from chromatography and spectroscopy is quite good in this regard.

With respect to the copolymer, the MALDI-TOF mass spectrum of fraction 1 gave no important additional information. Consequently the spectrum is not shown here.

In the case of fraction 2, however, MALDI-TOF-MS gave a very clear picture with respect to the MMDs and functionalities of the PBO homopolymers.

It is shown in Fig. 11 that two important series of ions were detected. The series with higher intensity is the PBO diol homopolymer, again ionized by attachment of Na⁺ (e.g. *m/z* = 1050, 1122, 1194 Da; the maximum of the MMD being around 1000 Da). The series with lower intensity – e.g. at *m/z* = 888, 960, 1032, 1104 Da – is an unambiguous proof for the PBO homopolymer with an unsaturated end group. Hence, these MALDI data verify and complete the chromatographic results.

In the case of MeO-PEO1100-*b*-PHO comparable results were obtained by MALDI-TOF-MS. The spectrum of the MeO-PEO1100-*b*-PHO raw sample is reproduced in Fig. 12. Again a complex pattern was observed for the copolymer due to the polydispersity of both blocks, but all peaks observed are assignable (as example see inset of Fig. 12). The MMD of the copolymer (mass range: 1300–2800; maximum at approximately 1950 Da) has the composition estimated from the relative intensity of assigned peaks corresponded again quite well to the chromatographic results. The PHO diol homopolymer was also detected as a series of [MNa]⁺ ions, e.g. at *m/z* = 1042, 1142, 1242 Da. After separation by LCCC MALDI-TOF of the corresponding fraction 2 (from Fig. 3) showed the PHO homopolymer products in more detail (see Fig. 13). PHO diols are the main product, while the PHO homopolymer with unsaturated end group was detected with rather low intensity (compare also inset of Fig. 13).

Summing up, the MALDI mass spectra verify and complete the results obtained by the different chromatographic approaches.

4. Conclusion

Polyoxyalkylene diblock copolymers can be fully characterized by combination of different chromatographic techniques and MALDI-TOF-MS. The complete characterization of polyoxyalky-

lene block copolymers that can reveal all required informations namely; MMD of both blocks in copolymer, MMD and functionality of homopolymer formed as a side product is not possible by using a single technique. The block copolymer was separated from homopolymer at CAP for hydrophobic block on semi-preparative scale and fractions were collected. These fractions were then analysed independently by SEC, LCCC, LAC and MALDI-TOF-MS in second dimension. This complete characterization would help in understanding the polymerization mechanism and sources of different side products, which in turn would help in better ideas about optimisation of reaction conditions and minimizing these side reactions. Furthermore the chromatographic results are in excellent agreement with that of spectroscopy.

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